THE STACK OF COHERENT SHEAVES IS LIMIT PRESERVING WHEN B -> S IS LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1308. Standing situation for the coherent stack.
- Definition artin-definition-limit-preserving (canonical tag 07XL). cited at statement line 1311. The property being established.
- Remark spaces-limits-remark-limit-preserving (canonical tag 05N0). cited at proof line 1330. Reduce to limits over B.
- Lemma spaces-limits-lemma-descend-modules-finite-presentation (canonical tag 07V7). cited at proof line 1339. The colimit identity for finitely presented modules.
- Lemma spaces-limits-lemma-descend-flat (canonical tag 08K0). cited at proof line 1346. Flat-over-base subcategory.
- Lemma spaces-limits-lemma-eventually-proper-support (canonical tag 08K2). cited at proof line 1347. Proper support subcategory.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line: the lemma is phrased in this situation; load-bearing.

Source reference. Original definition 07XL: sources/stacks-chow/tags.txt, source lines 10081–10081. Cited at statement line: the property being established; load-bearing.

Source reference. Original remark 05N0: sources/stacks-chow/tags.txt, source lines 7289–7289. Cited at proof line 1330: bottom arrow equivalence; reduce to limits over B.

Source reference. Original lemma 07V7: sources/stacks-chow/tags.txt, source lines 9997–9997. Cited at proof line 1339: colim FP(X_i) = FP(X_T).

Source reference. Original lemma 08K0: sources/stacks-chow/tags.txt, source lines 10868–10868. Cited at proof line 1346: flat subcategory.

Source reference. Original lemma 08K2: sources/stacks-chow/tags.txt, source lines 10870–10870. Cited at proof line 1347: proper support subcategory.

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Lemma 0.2. *In Situation 08KB assume that $B \to S$ is locally of finite presentation. Then $p : \mathcal{C}oh_{X/B} \to (Sch/S)_{fppf}$ is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Write $B(T)$ for the discrete category whose objects are the S-morphisms $T \to B$. Let $T = \lim T_i$ be a filtered limit of affine schemes over S. Assigning to an object $(T, h, \mathcal{F})$ of $\mathcal{C}oh_{X/B,T}$ the object h of $B(T)$ gives us a commutative diagram of fibre categories

$$\begin{array}{ccc} \operatorname{colim} \mathcal{C}oh_{X/B,T_i} & \longrightarrow & \mathcal{C}oh_{X/B,T} \\ \downarrow & & \downarrow \\ \operatorname{colim} B(T_i) & \longrightarrow & B(T) \end{array}$$

We have to show the top horizontal arrow is an equivalence. Since we have assumed that B is locally of finite presentation over S we see from Limits of Spaces, Remark 05N0 that the bottom horizontal arrow is an equivalence. This means that we may assume $T = \lim T_i$ be a filtered limit of affine schemes over B. Denote $g_i : T_i \to B$ and $g : T \to B$ the corresponding morphisms. Set $X_i = T_i \times_{g_i,B} X$ and $X_T = T \times_{g,B} X$. Observe that $X_T = \operatorname{colim} X_i$ and that the algebraic spaces X_i and X_T are quasi-separated and quasi-compact (as they are of finite presentation over the affines T_i and T). By Limits of Spaces, Lemma 07V7 we see that

$$\operatorname{colim} FP(X_i) = FP(X_T).$$

where $FP(W)$ is short hand for the category of finitely presented $\mathcal{O}_W$-modules. The results of Limits of Spaces, Lemmas 08K0 and 08K2 tell us the same thing is true if we replace $FP(X_i)$ and $FP(X_T)$ by the full subcategory of objects flat over T_i and T with scheme theoretic support proper over T_i and T. This proves the lemma. □